

Experiment Name: Implementation of a Simple Http Server to Response text.

Theory: A simple HTTP server is a program that receives requests from clients and sends responses using the HTTP protocol.

It works using client-server communication over a network.

The server listens on a specific port for incoming requests.

After receiving a request, it processes the data and returns a text response.

It helps in understanding networking and web server fundamentals.

Code:

```
import java.io.IOException;
import java.io.OutputStream;
import java.net.InetSocketAddress;
import com.sun.net.httpserver.HttpServer;
import com.sun.net.httpserver.HttpHandler;
import com.sun.net.httpserver.HttpExchange;

public class SimpleHttpServer {
    public static void main(String[] args) throws IOException {
        int port = 8080; // You can change the port if needed
        HttpServer server = HttpServer.create(new InetSocketAddress(port),
0);

        server.createContext("/", new MyHandler());
        server.setExecutor(null); // creates a default executor
        System.out.println("Server started at http://localhost:" + port);
        server.start();
    }

    static class MyHandler implements HttpHandler {
        @Override
        public void handle(HttpExchange t) throws IOException {
            String response = "Hello, this is a simple HTTP server
response!";

            t.getResponseHeaders().add("Content-Type", "text/plain");
            t.sendResponseHeaders(200, response.length());
            OutputStream os = t.getResponseBody();
            os.write(response.getBytes());
            os.close();
        }
    }
}
```

```
}
```

Output:

```
Hello, this is a simple HTTP server response!
```